3 Biotech (2023) 13:248
https://doi.org/10.1007/s13205-023-03666-0

ORIGINAL ARTICLE

Genome-wide identification and analyses of the AHL gene family in rice (*Oryza sativa*)

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Received: 27 September 2022 / Accepted: 11 June 2023 / Published online: 24 June 2023

Abstract

AHL (AT-HOOK MOTIF CONTAINING NUCLEAR LOCALIZED) family members play a critical role in stress resistance regulation by DNA–protein and protein–protein interactions in a number of plant biological processes. Using genomic data, an attempt was made to evaluate AHL genes in rice. Using a genome database, we performed in silico detection and characterization of AHL family genes in rice. The data of the gene were obtained from the Rice Genome Annotation Project (RGAP) database. The rice genome data were analyzed using bioinformatics software. The main objectives of the research are genome-wide recognition, expression, structural examination, phylogenetic analysis of AHL gene family, classification of AHL proteins into different classes based on motif and domain composition, analysis of promoter regions to identify stress and phytohormone-associated cis-elements, expression analysis of OsAHL genes in diverse tissues and stressful situations and understanding the roles of AHLs in controlling rice plant development. The genome-wide recognition, expression, and structural examination of the AHL gene family were undertaken in this research to evaluate the structural activities of AHLs in rice. From the *Oryza sativa* genome, 26 AHL genes have been identified. WoLF PSORT analysis predicted different subcellular localizations for these proteins, including nuclear, cytoplasmic, chloroplast, and endoplasmic reticulum. According to a phylogenetic study, rice AHLs resulted in two clades: Clade-A with no introns (excluding OsAHL15 and OsAHL21) and Clade-B with four introns. Depending on the AT-hook motif (s) (AHM) and PPC/DUF 296 domain composition, the AHL proteins are categorized into the following three classes: Type-I, Type-II, and Type-III, among Type-I AHLs constituting Clade-A, Type-II, and Type-III creating Clade-B. Type-I was the largest gene family, representing 57.69% of OsAHL genes. The exon–intron organization within clades of OsAHL genes was similar. Multiple sequence alignment identified 15 conserved motifs, including AT-hook motifs and the PPC domain, suggesting DNA-binding functionality. OsAHL genes were distributed across 12 chromosomes, with chromosome 2 and 8 harboring the highest number of genes. Gene duplication analysis revealed eight paralogous pairs, indicating evolutionary divergence between 13.32 and 35.59 million years ago. The emergence of OsAHL paralogous pairs was favored by purifying selection. Synteny analysis between rice and Arabidopsis demonstrated collinearity among AHL gene pairs, implying comparable structure and function in the two species. The role of stress- and phytohormone-associated cis-elements in the OsAHL genes was discovered by promoter analysis. OsAHL genes participated in various biological processes, with a prominent involvement in cellular and metabolic processes. They exhibited a significant enrichment in binding functions, including a substantial proportion of transcription regulators. OsAHL genes displayed diverse expression patterns in different tissues and under abiotic stress conditions. According to their expression patterns, the majority of OsAHLs of Clade-B were expressed mainly in the pistil indicating their roles in flower formation, while Clade-A OsAHLs had the minimal expression in pistil and highly expressed in embryos, indicating that the AHLs within each clade had the same expression patterns. Some OsAHL genes were also expressed in stressful situations, such as cold, salt, and drought. Protein interaction analysis revealed networks involving AHL proteins and other proteins, suggesting their participation in phytohormone responses, abiotic stress, and plant development. In this work, 26 OsAHL genes were found in the genome of rice. Rice OsAHLs were grouped into two phylogenetic groups. It is further divided into three types on the basis of the motif and domain composition. At various phases of development, the expression analysis of OsAHLs showed numerous variations in expression levels in diverse tissues and stress situations. Our findings shed light on the significant roles of AHLs in controlling rice plant development.

Extended author information available on the last page of the article

Keywords Rice · AHL family · Phylogenetic · AT-hook motif · Expression profiling

Introduction

The AHL family of transcription factors is found in every transcribed dicot and monocot land plant. Two domains were found to be conserved in this gene family: the first is the AT-hook motif, and the second is the plant and prokaryote domain [PPC, or Domain of Unknown Function 296 (DUF296)] (Zhao et al. 2013). The AT-hook motif binds to the minor groove of AT-rich sites [AA(T/A)T] that modifies DNA structure to regulate the expression of genes. It has an Arginine-Glycine-Arginine amino acid sequence at its core, followed by Arginine-Lysine or Proline sequence (Huth et al. 1997; Xiao et al. 2009). The conserved regions of AT-hook motifs differ by type, Type-I, and Type-II sequence (Zhao et al. 2013). AHL Type-I motifs contain the Gly-Ser-Lys-Asn-Lys consensus sequence, while the Arg-Lys-Tyr found at the C-terminal and the Arg-Gly-Argcentre sequence are found in AHL Type II motifs (Fujimoto et al. 2004). Increasing evidence indicates that AHL proteins have the AT-hook motif and the PPC/DUF296 conserved domain, while the bacterial proteins contain a 120-amino-acid domain (Fujimoto et al. 2004). This domain interacts with other transcriptional regulators and is required for nuclear localization (Fujimoto et al. 2004). Several researchers suggest that the PPC domain impacts the transcriptional activation of AHL proteins (Zhao et al. 2013). On the other hand, little evidence is available on this domain's significant role in the regulation of plant development. AHL genes are found in *A. thaliana* (Zhao et al. 2013) and several plants like *Zea mays* and *G. raimondii* (Bishop et al. 2020; Zhao et al. 2020). The role of AHL genes have been widely investigated in *A. thaliana* and shown to impact the developmental stages of the plant. Root growth, main root length, and lateral root density were all enhanced when AtAHL18 was overexpressed (Sirl et al. 2020). AtAHL4 is a homologous gene of AtAHL3, which governs the progress of vascular tissue boundaries in the roots of *A. thaliana* (Zhou et al. 2013). In the FLOWERING LOCUS, AtAHL22 regulates the flowering period by binding histone deacetylases (HDAC) to the AT-rich region (Yun et al. 2012). Inhibition of petiole development in A. thaliana is caused by the SOB3 gene (SUPPRESSOR OF PHYTOCHROME B4-#3), which encodes an AT-hook protein (Favero et al. 2020). AHL genes have also been linked to abiotic and biotic stress responses, and they control the expression of the gene of interest or bind with other proteins to contribute significantly to the development of plant and environmental stress actions. By reducing PAMP-triggered NHO1 and FRK1 expression, AtAHL20 suppresses pathogen-associated molecular patterns (PAMP)-induced responses (Lu et al. 2010). AtAHL10 is involved in the low-water potential stress response and interacts with the protein Highly ABA-Induced 1 (HAI1) (Wong et al. 2019). During the panicle growth stage, OsAHL1 improves rice drought resistance by favorably regulating drought-related genes. Furthermore, overexpression of OsAHL1 increased resilience to cold and salt stresses (Zhou et al. 2016).

Recently, AHL gene family has been identified and analyzed in several species. 29 AHL genes (AtAHLs) were identified from the Arabidopsis genome (Matsushita et al. 2007). The sorghum genome contains 22 AHL genes (Zhao et al. 2014). 37 AHLs were identified from the maize genome (Bishop et al. 2020). 48, 51, and 99 AHLs were identified from three cottons (*Gossypium raimondii, G. arboretum,* and *G. hirsutum*), respectively (Zhao et al. 2020). The soybean genome contains 63 AHL genes (Wang et al. 2021b).

About 4000 years ago, a natural hybridization event occurred between two wild rice species, *Oryza rufipogon* (AA, $2n=24$) and *Oryza nivara* (CC, $2n=24$), giving rise to *Oryza sativa* (AA, $2n=24$), commonly known as cultivated rice (Huang et al. 2012). *Oryza sativa* is one of the most important cereal crops and serves as a model organism for studying various biological processes. Rice productivity is often affected by a range of biotic and abiotic stresses. Although AHLs are expected to have significant roles in these stress responses, a comprehensive genome-wide analysis of the AHL gene family in *Oryza sativa* has not been conducted. In this study, we identified 26 AHL genes (OsAHLs) from the *Oryza sativa* genome and investigated their molecular evolution characteristics.

The main objective of the study is to identify and analyze the AHL gene family in rice (*Oryza sativa*) at a genome-wide level. The gap in the study is that there are very few previous studies on AHL gene family and no detailed study was on *O. sativa* which include cis-element analysis, upregulation of gene expression in stress conditions, chromosomal localization, conserved motif analysis, protein–protein interaction, etc. Through this research, the authors aim to expand our understanding of this gene family in rice and its potential biological significance. We aim to address through their analysis, such as size and composition of the AHL gene family in rice, AHL genes distribution across the rice genome, gene expression or functional enrichment among the AHL genes in rice, and the potential role of the AHL gene family in rice growth, development, and/or response to environmental stimuli. By outlining these hypotheses, we can help readers understand the specific goals of their study and how the results may contribute to the broader field of plant genetics and biology. By showing the possible functional variation of rice AHL genes, this investigation will contribute to our knowledge of the AHL gene family. The assessment of the OsAHLs' probable roles will be aided by an extensive genome-wide analysis based on a sequenced genome. The gene structure, subcellular localization, phylogenetic analysis, promoter, and synteny analysis, conserved motifs, expression levels, and protein–protein interaction of 26 AHLs found in *O. sativa* are all investigated. Our results will lay the groundwork for future research into the roles of AHLs in plants.

Materials and methods

Retrieval, subcellular localization, and physico-chemical features of AHL genes in *O. sativa*

The amino acid sequences of 29 AtAHL proteins were obtained from the TAIR (The Arabidopsis Information Resource; https://www.arabidopsis.org/) database (Zhao et al. 2014). These sequences were used to create a blast query in Phytozome (12.1; https://phytozome.jgi.doe.gov/pz/portal.html) using several parameters such as target type, proteome; program, BLASTP-protein query to protein database; and expect (E) threshold, 1. Duplicate sequences were removed. The SMART database (http://smart.embl-heidelberg.de/) was used to confirm that the conserved AT-hook motif and PPC domain were present in the putative OsAHL amino acid sequences (Letunic and Bork 2018). The gene sequence and coding amino acid sequence for every OsAHL gene were taken straight from Phytozome. WoLF PSORT (https://www.genscript.com/wolf-psort.html) was used to determine subcellular localization (Horton et al. 2007). Expasy ProtParam software (https://web.expasy.org/protparam/) was used to analyze the length of the protein, theoretical pI, aliphatic index, molecular weight, and grand standard of hydropathy (GRAVY) (Gasteiger et al. 2005).

Phylogenetic analysis

Using the OsAHL genes as a query, standard protein BLAST was used to find amino acid sequences from *A. thaliana* in the TAIR database (The Arabidopsis Information Resource; https://www.arabidopsis.org/). Multiple sequences were aligned using Clustal W and Bioedit (version 7.2.5); with gap opening (10) and gap extension penalties (0.1). The maximum likelihood (ML) approach was used to analyze the molecular characteristics and evolutionary associations of AHL genes in *A. thaliana*, *G. raimondii*, *Zea mays,* and *O. sativa,* with several parameters, such as 500 bootstrap replications, Poisson model, and partial deletion (95%) (He et al. 2013). Phylogenetic analysis allows researchers to infer molecular characteristics such as gene duplication events, evolutionary relationships, functional divergence, and conservation of sequence motifs or domains within the AHL gene family (Wang et al. 2021a, b). These insights contribute to understanding the structure, function, and evolution of the gene family and provide a foundation for further experimental investigations. Phylogenetic analysis of AHL genes in Arabidopsis can help elucidate their evolutionary relationships and functional diversification. By comparing the amino acid sequences of AHL genes and constructing phylogenetic trees, researchers can infer the evolutionary associations between different AHL gene family members (Zhao et al. 2014). This analysis can provide insights into gene duplication events, divergence, and the emergence of distinct clades or subgroups within the gene family. Additionally, molecular characterization of AHL genes in Arabidopsis involves studying their expression patterns, protein–protein interactions, and functional annotations. Techniques such as transcriptomic analysis, protein localization studies, and genetic manipulation experiments contribute to understanding the molecular characteristics and biological roles of AHL genes in Arabidopsis (Zhao et al. 2013).

Gene structure and conserved motif analysis

The organization of exon-introns in OsAHL genes was created with GSDS software (Gene Structure Display Server version 2.0: http://gsds.cbi.pku.edu.cn/). The conserved regions in OsAHL genes were identified using MEME (Multiple Em for Motif Elicitation software 5.1.1; http://meme-suite.org/tools/meme) software by the default parameters setting and a conserved motif number of 15.

Chromosomal localization and gene duplication

The chromosomal distribution of OsAHL genes was drafted by MapInspect software (http://www.plantbreeding.wur.nl/uk/software_mapinspect.html) according to their position information available in the Rice Genome Annotation Project (RGAP) database (http://rice.plantbiology.msu.edu/index.shtml). Duplicate gene pairs derived from the Plant Duplicate Gene Database (PlantDGD) in five modes of gene duplication, including whole-genome duplication (WGD), tandem duplication (TD), proximal duplication (PD), transposed duplication (TRD), and dispersed duplication (DSD). The programme TBtools were used to calculate the synonymous substitution rate (Ks) and non-synonymous substitution rate (Ka) to further analyze the divergence of duplicated genes (Chen et al. 2020). The divergence time (*T*) was subsequently calculated by the formula $T = \mathrm{Ks}/2\lambda \times 10^{-6}$ million years ago (Mya) based on a rate of λ ($\lambda = 6.5 \times 10^{-9}$ for rice) substitutions per synonymous site per year (Fan et al. 2021).

Synteny and collinearity analysis

Synteny analysis of protein family genes between the *O. sativa* and *A. thaliana* genomes. The GFF and genomic files of both species were retrieved from Phytozome (version 12.1; https://phytozome.jgi.doe.gov/pz/portal.html) and the conserved regions of the genes present in both species' genomes were identified using TBtools (Chen et al. 2020). The dual synteny analysis in TBtools or the one step McScan X tool to retrieve or present the evolutionarily conserved regions of the proteins in two species genomes. OsAHL duplications were found using the MCscanX software. For the protein sequences, BLAST was used with an e-value of 1.0e−5. Duplications were characterized as dispersed, singleton, WGD, segmental, and tandem duplications. The potential WGD or segmental duplications of OsAHL are represented in a schematic diagram with lines connecting them.

Promoter cis-element analysis

OsAHL gene promoters were obtained from Phytozome and used for cis-regulatory element analysis 1500 bp upstream of translation initiation. The cis-elements in each promoter were found using the PlantCARE database (Lescot et al. 2002). Using TBtools, a graphical representation of cis-regulatory elements found in the promoter region of OsAHL genes was created (Chen et al. 2020).

Expression analysis of OsAHL genes

The Rice Genome Annotation Project (RGAP) database (http://rice.plantbiology.msu.edu/index.shtml) was utilized to extract fragments-per-kilobase-per-million (FPKM) expression data to analyze the expression analysis of OsAHL genes, the development stages, and the stress conditions for the transcriptome data. The varieties used in experiments are sensitive varieties. Transcripts were constructed using Cufflinks software for each RNA-seq analysis submitted by Buell Lab, Department of Plant Biology, Michigan State University (MSU-BUELL), and digital gene expression submitted by Gene Expression Omnibus (GEO) (Trapnell et al. 2012). The heatmap charts were created based on gene expression levels (FPKM).

Gene ontology (GO) annotation

The following procedures were done to provide a functional characterization of OsAHL genes by Blast2GO in OmicsBox Version 2.0.36: UniProtKB/Swiss-Prot (swissprot_ v5) was used to align and map complete protein sequences. We used a 55-point annotation threshold, a GO weight of 5, and an *E* value-hit-filter of 1.0E−6 to annotate into the three GO categories (biological processes, molecular functions, and cellular components) (BioBam Bioinformatics 2019).

Protein interaction network analysis

The interacting network of OsAHL proteins in *Oryza sativa* was estimated using the STRING database (version 11.0; https://string-db.org/cgi/input.pl) (Szklarczyk et al. 2015). The sequence of every OsAHL protein was utilized as a query, and then *O. sativa* was chosen from the "Organism" choices. The "active interaction sources" were changed to "text mining," "experiments," "databases," "co-expression," "neighborhood," "gene fusion," and "co-occurrence," and the "meaning of network edges" was changed to "evidence." The minimum needed interaction score was set to 0.4 with a medium level of confidence. Finally, in Cytoscape, the interaction network of protein was generated and viewed (version 3.7.2).

Results

Retrieval, subcellular localization, and physico-chemical features of AHL genes in *O. sativa*

From the complete genome of *O. sativa*, we found 26 AHL genes. These AHL genes were given names based on their chromosome positions, ranging from OsAHL1 to OsAHL26. The molecular weight ranged from 20,510.56 to 143,190.39 Da, while the length of the protein ranged from 202 to 1274 aa. OsAHL21 was the largest of the proteins, whereas OsAHL15 was the smallest. The theoretical pI values of the OsAHL proteins varied significantly (5.45–10.46). The minimum GRAVY score was OsAHL9, while the maximum was OsAHL15 (Table 1). Nuclear localization was predicted for 18 proteins, cytoplasmic localization for seven proteins, and chloroplast localization for two proteins by WoLF PSORT. In the endoplasmic reticulum, OsAHL16 was predicted. Table 1 contains detailed information.

Phylogenetic analysis of OsAHL genes

The complete amino acid patterns of 26 AHLs from *O. sativa*, 29 from *A. thaliana*, 37 from *Z. mays* and 48 from *G. raimondii* were utilized to create an unrooted tree using the maximum likelihood (ML) approach to examine the ancestral connection. The MEGA 7.0 software was utilized to visualize the data. Three distinct types of AHL were identified: Type I, Type II, and Type III (Fig. 1). The PPC domain and the conserved sequences RGRPAGSKNKPKP and

Table 1 Basic information about AHLs in *O. sativa*

Sequence ID	Gene name	Chromosome	Location start–end	Strand	CDS (bp)	Protein (a.a.)	M.W[a] (Da)	pI[b]	GRAVY	The predicted location of OsAHL proteins
LOC_Os01g72450	OsAHL1	Chr1	42,012,736..42013581	Forward	798	266	26,638.08	7.87	− 0.123	Cytoplasmic
LOC_Os02g03270	OsAHL2	Chr2	1,313,909..1319031	Forward	1167	389	38,533.26	9.09	− 0.033	Nuclear
LOC_Os02g25020	OsAHL3	Chr2	14,507,846..14509480	Forward	951	317	32,170.76	6.03	− 0.309	Nuclear
LOC_Os02g48320	OsAHL4	Chr2	29,571,738..29574258	Reverse	1011	337	32,669.37	6.71	− 0.204	Nuclear
LOC_Os02g57520	OsAHL5	Chr2	35,242,144..35243851	Forward	801	267	26,625.91	5.76	− 0.236	Nuclear
LOC_Os02g57820	OsAHL6	Chr2	35,415,170..35419484	Reverse	1185	395	40,028.53	6.97	− 0.372	Nuclear
LOC_Os03g16350	OsAHL7	Chr3	9,022,589..9023976	Forward	777	259	25,858.91	6.60	− 0.219	Nuclear
LOC_Os04g50030	OsAHL8	Chr4	29,831,236..29833007	Reverse	918	306	30,691.22	6.18	− 0.248	Cytoplasmic
LOC_Os04g49990	OsAHL9	Chr4	29,814,032..29818148	Reverse	1140	380	38,240.73	9.06	− 0.479	Nuclear
LOC_Os04g58730	OsAHL10	Chr4	34,922,127..34926501	Reverse	1260	420	42,840.10	9.80	− 0.423	Nuclear
LOC_Os06g04540	OsAHL11	Chr6	1,962,245..1963580	Reverse	987	329	33,416.26	6.56	− 0.273	Nuclear
LOC_Os06g22030	OsAHL12	Chr6	12,758,357..12764136	Reverse	1137	379	37,450.18	10.46	− 0.224	Chloroplast
LOC_Os06g22100	OsAHL13	Chr6	12,825,418..12827565	Forward	969	323	32,009.64	5.45	− 0.262	Nuclear
LOC_Os06g41860	OsAHL14	Chr6	25,098,249..25099153	Forward	777	259	25,955.65	8.88	0.039	Nuclear, cytoplasmic
LOC_Os07g13100	OsAHL15	Chr7	7,501,889..7503889	Reverse	606	202	20,510.56	6.96	0.129	Cytoplasmic
LOC_Os08g02490	OsAHL16	Chr8	1,007,216..1011065	Forward	1119	373	37,813.56	9.42	− 0.120	Endoplasmic reticulum
LOC_Os08g06320	OsAHL17	Chr8	3,493,320..3494794	Reverse	870	290	28,298.74	6.29	− 0.045	Cytoplasmic
LOC_Os08g37345	OsAHL18	Chr8	23,589,602..23590577	Reverse	975	325	32,282.42	8.49	− 0.060	Cytoplasmic
LOC_Os08g40150	OsAHL19	Chr8	25,404,901..25409103	Reverse	1065	355	35,262.80	10.08	− 0.227	Nuclear
LOC_Os08g44910	OsAHL20	Chr8	28,201,023..28202721	Reverse	708	236	23,940.15	7.99	− 0.200	Nuclear, cytoplasmic
LOC_Os09g20040	OsAHL21	Chr9	12,004,418..12010546	Forward	3822	1274	143,190.39	5.85	− 0.172	Nuclear
LOC_Os09g28930	OsAHL22	Chr9	17,588,283..17589288	Reverse	1005	335	34,739.36	6.83	− 0.269	Nuclear
LOC_Os09g31470	OsAHL23	Chr9	18,964,579..18968444	Reverse	1080	360	35,491.25	10.15	− 0.165	Nuclear
LOC_Os10g42230	OsAHL24	Chr10	22,741,626..22745859	Reverse	1218	406	41,398.90	9.51	− 0.402	Nuclear
LOC_Os11g05160	OsAHL25	Chr11	2,271,005..2274083	Forward	1101	367	37,243.81	7.22	− 0.225	Nuclear
LOC_Os12g05200	OsAHL26	Chr12	2,305,430..2309611	Forward	843	281	29,392.51	9.82	− 0.176	Chloroplast

[a]Molecular weight of the amino acid sequence

[b]Isoelectric point

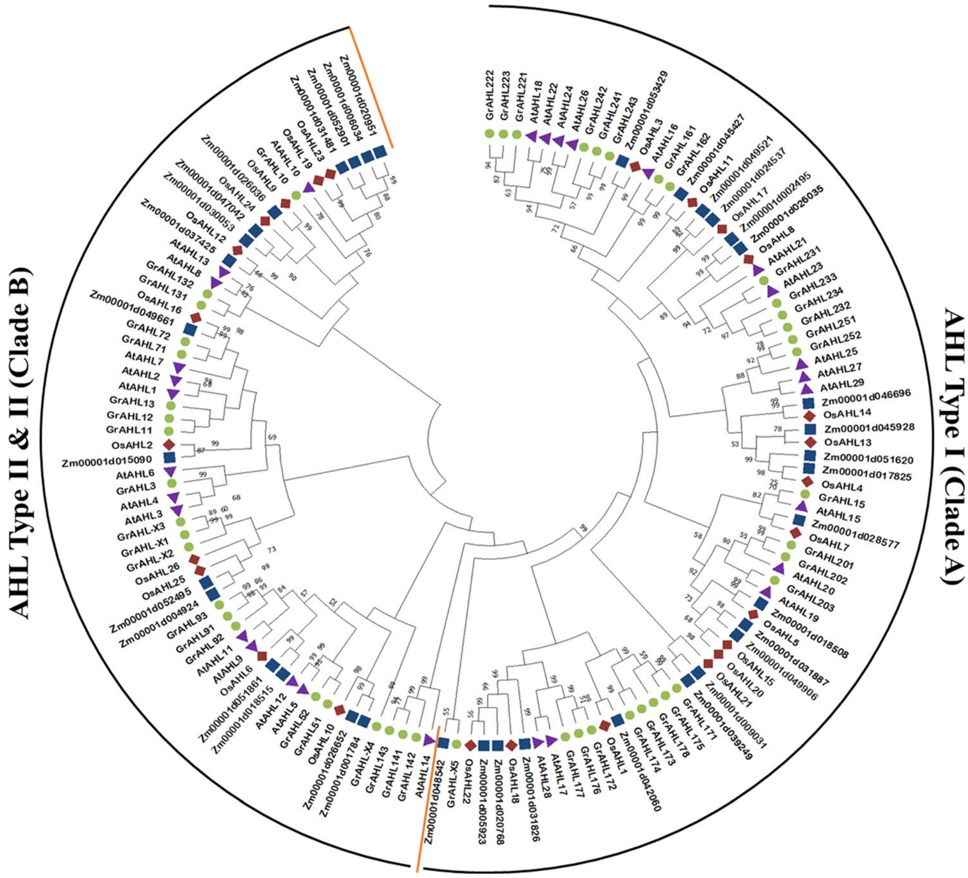


Fig. 1 Phylogenetic analysis of the AHL gene family members from *Oryza sativa* (Os), *Arabidopsis thaliana* (At), *Gossypium raimondii* (Gr), and *Zea mays* (Zm). The phylogenetic tree represents the evolutionary relationships among AHL gene family members from the four plant species. The branches with less than 20% bootstrap support have been collapsed for clarity. Different colors are used to represent the different species: *O. sativa* (red), *A. thaliana* (purple), *G. raimondii* (green), and *Z. mays* (blue). Two distinct groups, labeled as AHL Type-I and AHL Type-II and III, are observed within the phylogenetic tree. AHL Type-I group (Clade-A) differentiate from AHL Type-II and III group (Clade-B) by orange color. The phylogenetic tree was constructed using the Maximum Likelihood (ML) method implemented in MEGA7.0 software. The analysis was performed with 500 bootstrap replicates to assess the statistical support for the branching patterns observed in the tree. The phylogenetic analysis provides insights into the evolutionary relationships and classification of AHL gene family members in these plant species, highlighting the presence of distinct AHL gene groups and their divergence throughout evolution

RGRPPGSKNKPKP were found in Type I; the PPC domain and two AT-hook motifs (RGRPRKY and RGRP conserved sequences) were found in Type II, and the PPC domain and the RGRPRKY sequence were found in Type III. The AHL Type-I gene family has the most members, accounting for 57.69 percent of OsAHL genes, whereas AHL Type II and III genes contributed 42.30 percent. Arabidopsis AHL proteins were categorized as OsAHL proteins. These findings show that OsAHL proteins are highly correlated to proteins found in Arabidopsis.

Phylogenetic analysis, gene structure, and conserved motifs

We generated a phylogenetic tree of OsAHLs using MEGA7.0 and their corresponding protein sequences. Five paralogous pairings with significant bootstrap support (> 89%) were discovered (Fig. 2a). To investigate exon–intron organization, we compared the complete coding sequences of AHLs to their consequent genomic DNA sequences in *O. sativa*. Most OsAHL genes from the same clades, particularly paralogous pairs in the same group, have similar exon–intron lengths and numbers (Fig. 2b). The OsAHL Type-II and Type-III genes had four introns, whereas the OsAHL Type-I genes had only coding sequences (except OsAHL15 and OsAHL21) with upstream/downstream sequences (except OsAHL15, OsAHL21, OsAHL18, and OsAHL22).

To explore the existence of homologous domain sequences and the conservation degree in the two domains, AT-hook motif(s) and PPC domain, multiple sequence alignment was performed to build sequence logos of the

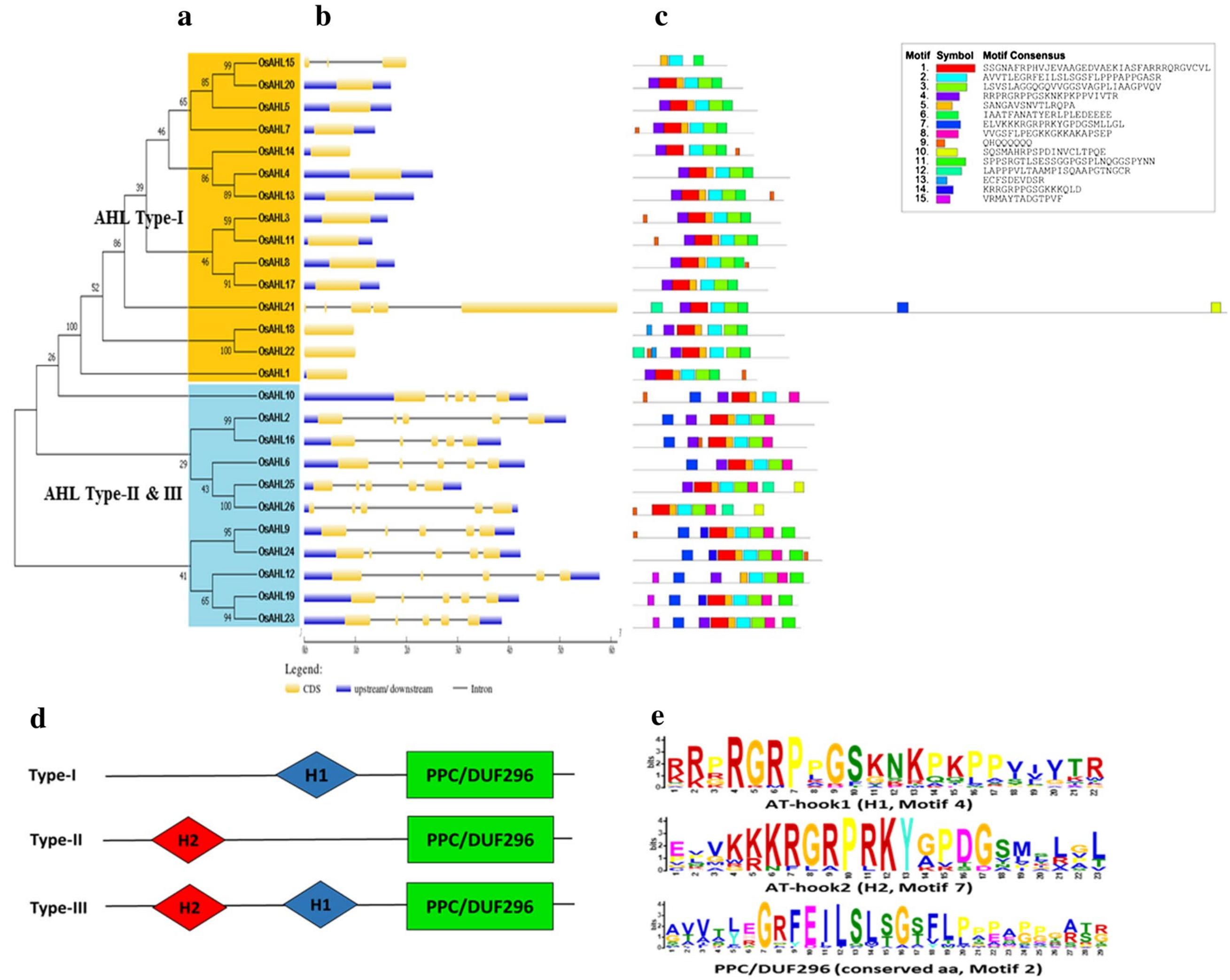


Fig. 2 Phylogenetic relationships and gene structure of AHLs in *O. sativa*. **a** Multiple alignment of full-length amino acid sequences of OsAHL genes was performed using Clustal W 2.0. The resulting phylogenetic tree was constructed using the Maximum Likelihood method implemented in MEGA7.0 software. AHL Type-I subfamily is represented by an orange background, while AHL Type-II and Type-III subfamily are represented by a blue background. **b** The gene structures, including the arrangement of exons and introns, of OsAHL gene family members are depicted. Exons are shown as boxes, and introns are represented by lines connecting the exons. The number of exons and their lengths may vary among different OsAHL genes. **c** Conserved motifs within the OsAHL protein sequences were identified using the MEME website. 15 distinct conserved motifs were discovered and are denoted by different colors. The sequence logo representation of the conserved motifs is displayed, providing a graphical depiction of the conservation patterns. **d** The topology of three types of AHL proteins in rice is illustrated based on the presence of AT-hook motifs and the PPC/DUF296 domain (motif 2). H1 represents the presence of AT-hook1, containing the conserved peptide found in motif 4, while H2 represents the presence of AT-hook2, containing the conserved peptide in motif 7. **e** The sequence logo visualization is presented to highlight the relative conservation of amino acid residues within the identified conserved motifs. The height of the letters in the sequence logo indicates the degree of conservation, with taller letters representing more highly conserved residues. The combination of phylogenetic analysis, gene structure depiction, identification of conserved motifs, and analysis of protein topology provides a comprehensive understanding of the AHL gene family in *O. sativa*, including their evolutionary relationships, gene organization, conserved motifs, and protein features

two domains in rice beside the MEME website. There were 15 conserved motifs predicted, each with its amino acid sequence (Fig. 2c, Supplementary File 1). AHL proteins were classified into three types (Type-I/II/III) based on the number and makeup of AT-hook motif(s) and PPC/DUF 296 domain, with Type-I AHLs comprising clade-A and the other two kinds branching in clade-B. AHL proteins have two kinds of AT-hook motifs (H1 and H2) (Fig. 2d). Both H1 and H2 in the AHL proteins have an R-G-R-P region, indicating that they can attach AT-rich B-form DNA (in the minor groove). H2, which has a prolonged region of R-G-R-P-R-R-R-K-Y heptapeptides, has greater conservation than H1 in rice. Clade A contained just H1, whereas clade B had H2 or H1 plus H2 (Fig. 2e). Two AT-hook motifs were found in the conserved structures of OsAHL2, OsAHL6, OsAHL10, OsAHL12, OsAHL16, OsAHL21, and OsAHL23, demonstrating a unique role in development. Excluding OsAHL15, almost all AHL genes in rice feature an AT-hook motif(s) and a PPC/DUF296 domain. Because OsAHL15 possessed the most PPC/DUF296 domain (motif 2) while missing the AT-hook motifs (motif 4 and motif 7), it was identified as a pseudogene in rice and utilized as an AHL family member for future study. Motifs 1 and 3 are also Plants and Prokaryotes Conserved (PCC) domain which has a similar function to motif 2. This is a PPC domain found in proteins that contain AT-hook motifs which strongly suggest a DNA-binding function for the proteins as a whole. Proteins with PPC domains are found in bacteria, archaea, and the plant kingdom (Fujimoto et al. 2004). Except these, no other motifs have known function.

Chromosomal locations and gene duplication

The distributions of OsAHL genes were determined by mapping on 12 chromosomes. Five OsAHL genes (OsAHL2, 3, 4, 5, and 6) were discovered on chromosome 2, and five more (OsAHL16, 17, 18, 19, and 20) were discovered on chromosome 8 (Fig. 3). On chromosome 6, four OsAHL genes were detected, and three were on chromosomes 4 and 9. On chromosomes 1, 3, 7, 10, 11, and 12, OsAHL1, 7, 15, 24, 25, and 26 were found, correspondingly. There were no genes discovered on chromosome 5. To further recognize the evolutionary boundaries of the OsAHL genes, we utilized TB tools to calculate the Ka/Ks values. The Ka/Ks ratio was lower than one in all OsAHL genes, demonstrating that most of the paralogous pairings developed under purifying selection (Supplementary File 2). The divergence date for these 8 paralogous pairs was calculated to be between 13.32 and 35.59 Mya (Supplementary File 2). Purifying selection typically acts to remove deleterious mutations and maintain functional genes within a population. In the context of gene duplication events leading to the emergence of paralogs, purifying selection generally acts to preserve the original gene's function and ensures its continued essential role in an organism.

When a gene duplication occurs, one of the duplicate copies (paralog) is often free from selective pressure and can accumulate mutations without detrimental effects. This can potentially lead to the loss of function or nonfunctionalization of one of the paralogs over time. However, in some cases, the paralogs may acquire distinct functions through a process called neofunctionalization or subfunctionalization (Innan and Kondrashov 2010).

Purifying selection plays a role in the emergence of paralogous pairs by preserving the functional integrity of both copies. While the duplicate gene initially may not have a specific function, it is likely to retain a similar structure to the original gene. This structural similarity allows the duplicate gene to maintain interactions with other molecules or participate in regulatory networks, providing a foundation for potential functional diversification (Lynch and Conery 2000).

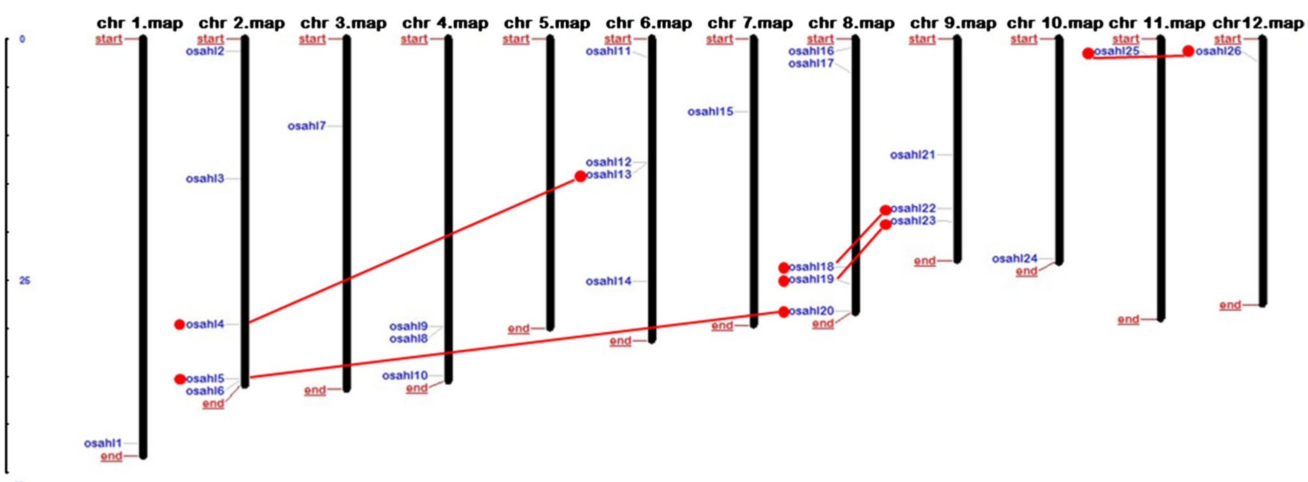


Fig. 3 Distribution of OsAHL genes on rice chromosomes. The figure displays the distribution of OsAHL genes on the twelve chromosomes of rice (Chr 1–Chr 12). The chromosome numbers are indicated at the top of each chromosome. The OsAHL genes are represented by labeled markers along the length of the chromosomes, indicating their respective locations on the chromosomes. The values following the OsAHL genes denote their specific positions on the chromosomes. Additionally, the figure highlights the presence of whole-genome paralogous pairs of OsAHL gene family members, which are connected by red lines. These red lines indicate the gene duplication events within the OsAHL gene family. The distribution pattern of OsAHL genes on rice chromosomes provides insights into their genomic organization and the occurrence of gene duplication events, contributing to the expansion and diversification of the OsAHL gene family in the rice genome

Purifying selection acts to remove mutations that would disrupt the basic structure or essential functions of the duplicate genes. Any mutations that arise and confer a detrimental effect on the fitness or function of the duplicates are likely to be eliminated by purifying selection. The duplicates that maintain their structural integrity and are not deleterious to the organism are more likely to persist and undergo further evolution (Innan and Kondrashov 2010).

In summary, while the emergence of paralogs through gene duplication events may initially be driven by neutral processes, purifying selection subsequently acts to preserve and maintain functional integrity of the duplicates. This allows for the potential neofunctionalization or subfunctionalization of paralogs, leading to the evolution of new gene functions while retaining the essential functions of the original gene.

Synteny and collinearity analysis

The collinearity of the AHL gene family in rice was found using BLASTP and the MCScanX (Multiple Collinearity Scan) approaches to determine a probable association between the AHL genes and potential duplication events. We created comparative syntenic maps of Oryza sativa linked with Arabidopsis thaliana to study the potential evolutionary pathways of the rice AHL gene family (Fig. 4). Finally, four collinear AHL gene pairs were discovered in rice and Arabidopsis. Because the synteny conservation between the rice and Arabidopsis genomes is great, this finding may indicate that the OsAHL and AtAHL genes in rice have similar structures and functions.

Promoter cis-element analysis

PlantCARE was used to explore stress- and phytohormone-connected cis-elements in the region of the promoter of OsAHL genes to better recognize how they are regulated after abiotic stress and phytohormone treatment. There were 13 phytohormones and stress-related cis-elements found (Fig. 5, Supplementary File 3). Except for OsAHL3, all of the genes 5, 6, 8, 9, 17, 23, and 25 have an ABA-responsive element (ABRE). OsAHL7 has a total of 10 ABREs. Except for OsAHL5, 15, 17, 20, 21, 23, 24, 25, and 26, the other 17 genes contained a methyl jasmonate (MeJA)-responsive element (CGTCA/TGACG), and 10 more OsAHLs had a salicylic acid-responsive TCA element. OsAHL1, 8, and 21 each have one GARE motif that performs as a gibberellin-reactive element. TGA (auxin-responsive element) was identified in the OsAHL3, 7, 12, 14, 17, and 25 genes, while AuxRR-core (auxin-responsive element) was identified in the OsAHL4, 13, 15, 18, and 23 genes. Additionally, 6 members had a P-box (gibberellin-responsive element), 4 members had a TATC-box (gibberellin-responsive element), and 7 contained TC-rich repeats. 17 OsAHL genes included cis-elements linked to abiotic stress, such as the low-temperature-responsive element (LTR) and the drought-inducible element (MBS). Both OsAHL3 and OsAHL19 have two LTR cis-elements. None of the 26 genes, on the other hand, identified any cis-elements linked to phytohormones or stress.

Gene ontology (GO) annotation

OsAHL genes play an important role in a variety of biological activities, according to a GO study (Fig. 6). Cellular and metabolic processes have the most OsAHL genes, followed by biological process regulation and biological regulation.

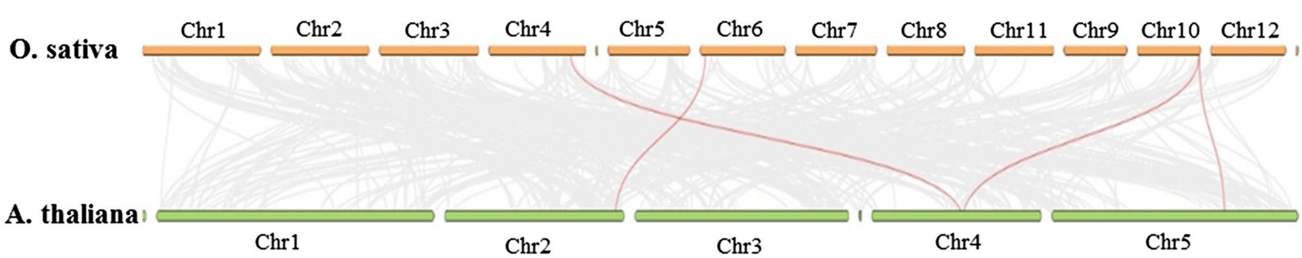


Fig. 4 Gene duplication and synteny analysis of AHL genes between *O. sativa* and *A. thaliana*. The figure depicts the gene duplication and synteny analysis between the AHL genes in *O. sativa* and *A. thaliana* genomes. In the background, grey lines represent collinear blocks within the rice and Arabidopsis genomes, indicating regions of conserved gene order and orientation. These collinear blocks suggest evolutionary relationships and shared ancestry between the two species. The red lines highlight the syntenic AHL gene pairs, indicating the presence of orthologous AHL genes between *O. sativa* and *A. thaliana*. These syntenic AHL gene pairs exhibit conserved gene positions and likely share functional similarities. The gene duplication events within the AHL gene family are demonstrated by the presence of duplicated gene pairs within both species. These gene duplication events contribute to the expansion and diversification of the AHL gene family in both *O. sativa* and *A. thaliana* genomes. The gene duplication and synteny analysis provide valuable insights into the evolutionary relationships, shared ancestry, and conservation of AHL genes between *O. sativa* and *A. thaliana*, shedding light on the functional conservation and divergence of AHL genes across these two plant species

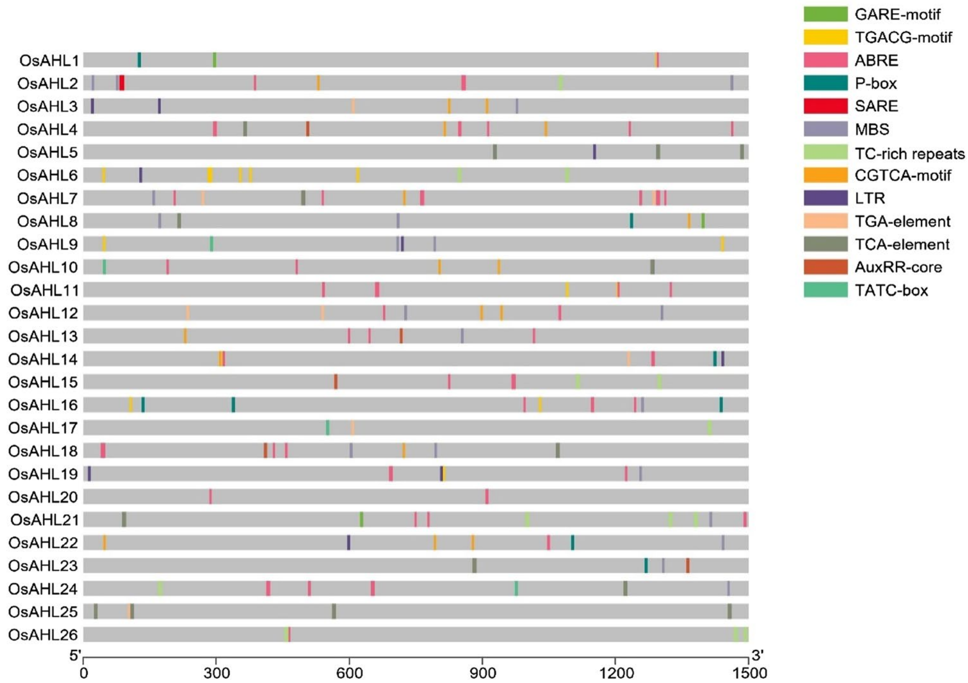


Fig. 5 Abiotic and phytohormone-related cis-elements in OsAHL gene promoters. The figure displays the location of cis-elements within the promoters of OsAHL genes, which are involved in the regulation of gene expression under abiotic stress conditions and phytohormone signaling. The visualization of cis-elements in each OsAHL gene was performed using TBtools software. Different colors are used to represent different cis-elements, highlighting their presence and distribution within the promoter regions. Abiotic stress-related cis-elements, such as ABRE (ABA-Responsive Element), and LTR (Low-temperature Responsive Element), are shown in specific colors, indicating their locations within the promoter sequences. Phytohormone-related cis-elements, including TGA-element (Auxin-responsive element), TCA-element (Salicylic acid-responsive element), GARE-motif (Gibberellin-responsive element), and others, are also displayed with distinct colors, reflecting their positions within the OsAHL gene promoters. The identification and visualization of these cis-elements provide insights into the potential regulatory mechanisms underlying the response of OsAHL genes to abiotic stresses and phytohormone signaling pathways. The analysis of abiotic and phytohormone-related cis-elements in OsAHL gene promoters contribute to a better understanding of the transcriptional regulation and functional roles of OsAHL genes in response to environmental cues and hormonal signals in rice

Reproduction, multicellular organismal processes, developmental processes, and reproductive processes accounted for 5.48% of the AHL genes. All 26 OsAHL genes were functionally enriched to bind for molecular function. Furthermore, 35 percent of OsAHL genes are transcription regulators. According to cellular component prediction, all OsAHL genes were discovered in cellular anatomical entities. Supplementary file. 4 has more information about GO annotation.

Gene expression profiles of OsAHLs

We examined the expression pattern of several AHL genes in O. sativa based on RNA-seq data collected from the Rice Genome Annotation Project (RGAP) database (http://rice.plantbiology.msu.edu/index.shtml) to investigate the putative biological activities of AHLs. *O. sativa* AHL genes were expressed in a variety of temporal and spatial patterns. The mostly OsAHL genes of Clade A (like AHL4, AHL7, AHL8, AHL11, and AHL13) have high expression levels in the embryo, but comparatively low expression levels in the endosperm, leaf, anther, and pistil. Some Clade-B OsAHL genes (like AHL6, AHL9, AHL10, AHL19, AHL23, and AHL24) exhibited substantial expression action in a variety of tissues, with the pistil being the most strongly expressed, showing that these genes have a particular function in flower formation (Fig. 7). The same expression pattern of AHLs within each clade was revealed in the expression result; however, the expression patterns of AHLs in one monophyletic clade differed from those in the other.

The expression patterns of different AHL genes in *O. sativa* are based on the GEO data retrieved from the Rice Genome Annotation Project (RGAP) database (http://rice.plantbiology.msu.edu/index.shtml) (Fig. 8). Out of 26 OsAHL genes, 19 OsAHL genes (except Clade A genes: AHL1, AHL3, AHL5, AHL11, AHL14, AHL18, and AHL22) showed expression activity in mature roots, and 16 OsAHL genes showed expression in mature pollen (Fig. 8). No OsAHL genes showed expression activity in drought-stressed leaves, and mature stigma, and ovary. In cold stress,

Fig. 6 Gene ontology (GO) analysis of OsAHL gene family. The figure presents the results of the Gene ontology analysis conducted on the OsAHL gene family, which provides insights into their predicted involvement in various biological processes, molecular functions, and cellular components. **a** Biological processes: The GO analysis predicts the biological processes in which OsAHL genes are likely to participate. The identified biological processes are represented by labeled nodes or terms in the figure. These processes may include but are not limited to cellular processes, metabolic processes, response to stimuli, developmental processes, reproductive process, and other key biological activities. **b** Molecular functions: The GO analysis also predicts the molecular functions of OsAHL genes. Molecular functions represent the specific biochemical activities or tasks performed by the gene products. The molecular functions are indicated by labeled nodes or terms in the figure, encompassing various molecular activities such as binding and transcription regulator activity. **c** Cellular components: The GO analysis further provides insights into the cellular components where OsAHL genes are likely to be located or function. Cellular components represent the specific structures or compartments within a cell. The identified cellular components are depicted as labeled nodes or terms in the figure, which include cellular anatomical activity. The GO analysis of the OsAHL gene family allows for a comprehensive understanding of the predicted roles and functions of these genes in different biological processes, molecular activities, and cellular locations. It provides valuable information about their potential contributions to the development, growth, and response of rice plants to various stimuli and environmental conditions

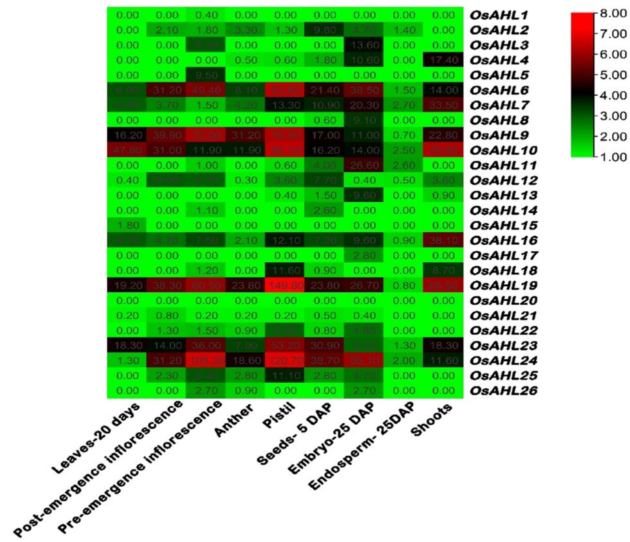


Fig. 7 Expression profiles of OsAHL genes. The figure presents a heatmap depicting the expression profiles of OsAHL genes based on RNA-seq data retrieved from the http://rice.plantbiology.msu.edu/index.shtml database. The heatmap represents the relative expression levels of OsAHL genes across different tissues, developmental stages, or experimental conditions. Each row corresponds to an individual OsAHL gene, while each column represents a specific sample or condition. The color scale displayed at the right side of the figure indicates the expression levels of OsAHL genes. Higher expression levels are represented by red color, indicating stronger gene expression, while lower expression levels are depicted in green color, indicating relatively lower expression. The heatmap provides a visual representation of the expression patterns and dynamics of OsAHL genes across different biological contexts. It allows for the identification of genes that are highly expressed or differentially expressed in specific tissues, developmental stages, or under particular experimental conditions. The analysis of expression profiles of OsAHL genes provides valuable insights into their potential roles and functions in different biological processes

OsAHL7, 8, 10, 23, and OsAHL19, 24 showed expressions in leaves and roots respectively. OsAHL4, 5, 23, and 24 genes showed expression in drought-stressed roots. Under salt stress, OsAHL20 and OsAHL7 showed expression in leaves and roots, respectively (Fig. 8). The results demonstrated that the OsAHLs genes regulate target gene expression, which is important for plant development and abiotic stress responses.

Analysis of the AHL protein interaction network in *O. sativa*

Using the protein sequences of O. sativa AHLs, we generated the protein interaction network using STRING. We observed a total of 25 protein interaction networks, except OsAHL21 (Fig. 9), representing that OsAHL proteins interrelate with the other proteins in reaction to phytohormone action and abiotic stress and throughout growth and development in *O. sativa*. We discovered that homologous proteins,

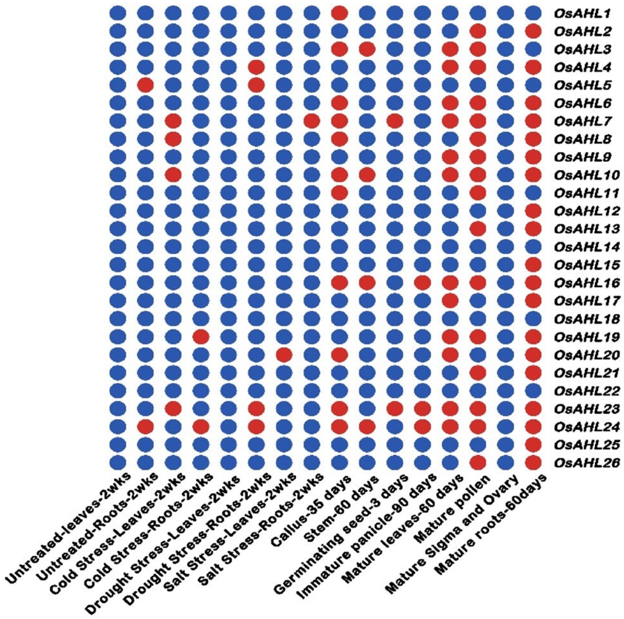


Fig. 8 Expression profiles of OsAHL genes. The figure presents a heatmap illustrating the expression profiles of OsAHL genes based on GEO (Gene Expression Omnibus) data retrieved from the http://rice.plantbiology.msu.edu/index.shtml database. The heatmap represents the relative expression levels of OsAHL genes across different experimental conditions, treatments, or time points. Each row in the heatmap corresponds to an individual OsAHL gene, while each column represents a specific sample or condition. Higher expression levels are represented by red color, indicating stronger gene expression, while lower expression levels are depicted in blue color, indicating relatively lower expression. The heatmap allows for the visualization and comparison of expression patterns among OsAHL genes under different experimental contexts. It facilitates the identification of genes that are highly expressed or differentially expressed in response to specific treatments or conditions. By analyzing the expression profiles of OsAHL genes, valuable insights can be gained into their potential roles and functions in various biological processes, their regulation in response to different stimuli, and their potential contributions to the growth, development, and stress responses of rice plants

such as OsAHL5 and OsAHL15, OsAHL8 and OsAHL17, OsAHL18 and OsAHL22, OsAHL19 and OsAHL23, and OsAHL4 and OsAHL13, have similar protein interaction networks, demonstrating that each protein pair has a conserved protein interaction domain. The OsAHL gene interacts with several proteins, and most of these proteins have known functions. Few of the protein functions are mentioned here Example: 1. SL1 protein, Zinc finger protein; Regulates floral organ identity and cell proliferation in the inner floral whorls. Probably specifies the identities of lodicule and stamen through positive regulation of MADS16 expression. May contribute to morphogenesis by suppressing OSH1 expression in the lateral organs. 2. PRR95; Two-component response regulator like; Controls photoperiodic flowering response. Seems to be one of the components of the circadian clock. 3. ELP3 protein: Elongator complex protein 3; Histone acetyltransferase. Part of the large multiprotein complex Elongator that is involved in the regulation of transcription initiation and elongation. May also have a methyltransferase activity. 4. SERR protein, Serine racemase; Catalyzes the synthesis of D-serine from L-serine. Has dehydratase activity towards both L-serine and D-serine. 5. CBT or Os07g0490200 protein; Putative calmodulin-binding protein; Transcription activator that binds calmodulin in a calcium- dependent manner in vitro. 6. OS09T0281700-01 or Os09g0281700 protein; May play an important role in plant growth and development. 7. TPP3 protein, Probable trehalose-phosphate phosphatase 3; Removes the phosphate from trehalose 6-phosphate to produce free trehalose. Trehalose accumulation in plant may improve abiotic stress tolerance. 8. TCTP protein, Translationally-controlled tumor protein homolog; Involved in calcium binding and microtubule stabilization. 9. OsJ_04190 protein, Uricase; Catalyzes the oxidation of uric acid to 5- hydroxyisourate, which is further processed to form (S)-allantoin; Belongs to the uricase family. 10. OsJ_28198 protein, Putative ripening regulated protein DDTFR18; Belongs to the multi antimicrobial extrusion (MATE) family. (STRING database (version 11.0; https://string-db.org/cgi/input.pl) (Szklarczyk et al. 2015).

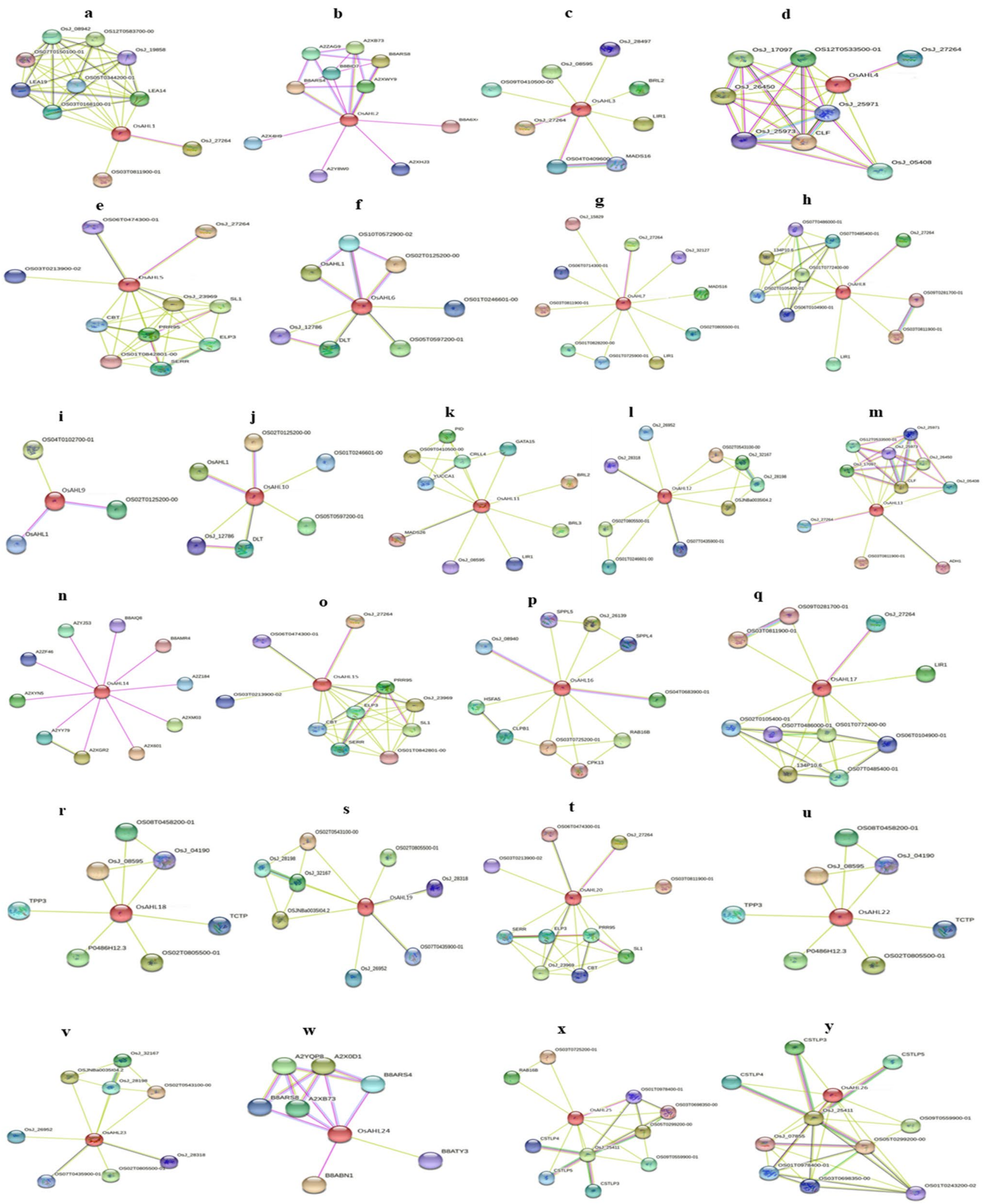
Discussion

The AHL gene family in Oryza sativa (rice) has been studied in the context of its crucial functions at different growth stages and stress responses. Previous research on AHL genes has primarily focused on *Arabidopsis*, *P. patens*, and other plant species, both monocotyledonous and dicotyledonous. However, in the present investigation, AHL genes were identified and characterized in *O. sativa* to understand their roles in this extensively farmed crop, which serves as a staple food for billions of people worldwide.

In the present study, the subcellular localization of OsAHL5 and OsAHL15, in the nucleus and cytoplasm respectively, suggests that these genes may have different functions and signaling pathways. The distinct subcellular localization of OsAHL18/22 further supports the idea that homologous genes can have diverse roles within a cell. AHL proteins, including the OsAHL proteins mentioned, have previously been detected in the nucleus in other studies. This localization suggests their involvement in nuclear functions such as transcriptional regulation and DNA repair. The presence of OsAHL14 and OsAHL20 in both the nucleus and cytoplasm suggests that these proteins may shuttle between these compartments, potentially participating in cellular processes that require coordination between the nucleus and cytoplasm. In addition to the nucleus, OsAHL proteins have been found in other cellular components, including the cytoplasm, chloroplasts, and endoplasmic reticulum (ER). This diverse localization indicates that OsAHL proteins likely have multiple roles and functions within the cell. The presence of several OsAHL proteins in the cytoplasm suggests their involvement in cytoplasmic signaling pathways. They may participate in processes such as signal transduction, protein synthesis, cytoskeletal organization, or other cytoplasmic events. The presence of OsAHL proteins in chloroplasts suggests their involvement in chloroplast-related functions. Chloroplasts are involved in photosynthesis, lipid metabolism, amino acid synthesis, and other important processes in plant cells. OsAHL proteins localized in chloroplasts may play a role in regulating these processes or mediating signaling between the chloroplast and other cellular compartments. The ER is involved in protein synthesis, folding, and transport within the cell. The localization of OsAHL proteins in the ER suggests their potential role in regulating ER-associated processes, such as protein quality control, lipid metabolism, or ER stress response. Overall, the subcellular localization of the OsAHL proteins provides valuable insights into their potential functions within different cellular compartments. The AT-hook motif and the PPC/DUF296 domain have both been proven to have a role in nuclear localization in the past (Sgarra et al. 2006; Street et al. 2008). As a result, we believe the existence of additional sequences has an impact on localization. The majority of AtAHL1 is located in the nucleoplasm, with a little quantity in the nucleolus (Fujimoto et al. 2004). Further studies are required to investigate the specific roles of these proteins in each location and to elucidate the signaling pathways in which they are involved.

Conservation of AHL gene family

Plant-specific AT-hook motif and PPC/DUF domain architecture are unchanged in the AHL gene family. The AT-hook motif and PPC motif sequences, as well as the gene sequence, gene structure, and motif count, differ amongst AHL family members. Using sequence logo analysis, the diversity of the AT-hook motif and the PPC domains in AHL proteins was explored further. AT-hook motif is differentiated by the evolutionary association of its homeodomains (s). Our findings revealed that rice AHL proteins with an enlarged core sequence R-G-R-P, notably in type II AT-hook motif, had a longer and more conserved core R-G-R-P-R-K-Y heptapeptide. Based on the AT-hook motifs and PPC domain, rice AHL proteins were grouped into three types, which corresponds to the previous study (Zhao et al. 2014; Huth et al. 1997). In the rice AHL genes, two forms of gene structure were discovered: intronless and multiple introns. Clade A, 13 OsAHL genes had no introns, whereas Clade-B genes had five exons. Furthermore, in *O. sativa*, the majority of AHL genes in Clade-B were primarily expressed in the pistil, with just minor expression in other organs. Clade-B OsAHL genes with numerous introns have been found in a variety of organs and tissues, implying that gene structure may influence gene expression patterns. Gene structure plays a crucial role in influencing gene expression patterns. The term "gene structure" refers to the organization and arrangement of various elements within a gene, such as exons, introns, promoters, enhancers, and other regulatory regions. These structural elements can directly or indirectly impact the expression of a gene. Here, Promoters are DNA regions located upstream of a gene that serves as binding sites for transcription factors. The structure of promoters, including the presence of specific DNA motifs or sequences, determines the binding affinity of transcription factors. Variations in promoter structure can lead to differences in gene expression levels. The putative OsAHL genes exhibited diverse expression patterns in various tissues. Some genes, such as AHL4, AHL7, AHL8, AHL11, and AHL13, showed high expression levels in the embryo but comparatively low expression in the endosperm, leaf, anther, and pistil. On the other hand, genes like AHL9, and AHL10 displayed substantial expression activity in multiple tissues, with the pistil showing the highest expression level. These expression patterns suggest potential roles for these genes in flower formation. The expression analysis of the OsAHL genes in response to abiotic stress, including drought, cold, and salt stress. For instance, AHL4, AHL5, AHL23, and AHL24 genes were found to be expressed in drought-stressed roots, indicating their potential involvement in the plant's response to drought stress. Similarly, specific genes like AHL7, AHL8, AHL10, and AHL19 showed expression in leaves or roots under cold stress. OsAHL genes regulate target gene expression and play important roles in plant development and abiotic stress responses, it implies that the putative OsAHL genes are likely involved in various biological processes related to plant growth, floral transition, and stress responses. Previous published works, such as those by Zhao et al. (2014, 2013) and Jin et al. (2011), have explored the roles of AHL genes in these processes, providing insights into the functional implications of AHL gene expression in plants. AHL genes within the same clade exhibited similar expression patterns, but those in different clades showed distinct expression patterns. This observation aligns with the concept of gene families or clades having shared expression profiles or regulatory mechanisms. Previous studies that have investigated AHL gene families or gene regulatory networks in rice or other plant species may provide further insights into the significance of these clade-specific expression patterns (Wang et al. 2021a, b). Overall, to establish stronger connections between the expression of the putative OsAHL genes and previously published works, it would be beneficial to refer to specific studies such as those by Zhao et al. (2014, 2013) and Jin et al. (2011), and



explore additional literature that focuses on the expression, function, and regulatory networks of AHL genes in rice and other plant species (Zhang et al. 2021; Zhao et al. 2020).

◂**Fig. 9** Predicted protein interaction network of OsAHL proteins (**a**–**y**), The figure presents the predicted protein interaction network of OsAHL proteins (**a**–**y**) based on analysis performed using the STRING database. Each node in the network represents an OsAHL protein, labeled as **a**–**y**, indicating individual members of the OsAHL gene family. The lines connecting the nodes represent the potential protein–protein interactions between OsAHL proteins. Different colored lines indicate different evidence of interaction, reflecting the reliability or confidence of the predicted interactions. The color code for the lines may vary depending on the specific STRING database settings used, but typically, it represents the strength or quality of the interaction evidence. The predicted protein interaction network provides insights into the potential associations and functional relationships between OsAHL proteins. It suggests potential partners or interactors that may contribute to the regulation, signaling, or physical interaction with OsAHL proteins, providing a basis for further investigation into their molecular functions and biological roles. It is important to note that the predicted protein interactions should be further experimentally validated to confirm their existence and functional relevance in the context of OsAHL proteins

The promoter analysis and protein interaction of OsAHLs

All OsAHL family members have at least a phytohormone-linked or stress-linked cis-element, specifying their function in response regulation. We also discovered LTR and MBS cis-elements in a number of OsAHL promoters, including OsAHL 3, 5, 6, 9, 14, 19, 22 and OsAHL2, 3, 7, 8, 9,12, 13, 16, 18, 19, 21,22,23, 24 respectively. Understanding the involvement of orthologous proteins in biological processes is possible due to the protein interaction network (Hao et al. 2016). The protein interaction network of OsAHLs was predicted using STRING. The homologs of Type II and III subfamilies, OsAHL18/OsAHL22 and OsAHL19/OsAHL23 shared a comparable protein interaction network, confirming their importance under low temperature and water stress, according to our networks. AHLs in *O. sativa* play an important part in the plant's response to environmental stressors and phytohormone treatments, according to these studies. We identified specific cis-elements, such as LTRs (Long Terminal Repeats) and MBSs (MYB-binding sites), in the promoters of the putative OsAHL genes. However, this indicates that we provided fewer insights into how these cis-elements function under the selected abiotic stress conditions and their correlation with the putative genes or stress tolerance. The presence of specific cis-elements in gene promoters suggests their potential involvement in gene regulation and response to stress conditions. For example, MBSs are known to be associated with drought-responsive gene expression, while LTRs are often associated with retrotransposon elements that can influence gene regulation. To determine how these cis-elements contribute to stress tolerance or gene expression under abiotic stress conditions, we typically conduct the promoter analysis. This analysis can provide insights into the regulatory mechanisms involving cis-elements, transcription factors, and target gene expression. The correlation between the identified cis-elements and their putative genes can be established by studying the expression patterns of the genes under stress conditions and assessing the presence or activity of the cis-elements in their promoters. Under abiotic stresses, promoter cis-elements such as LTR and MBS cis-element play pivotal roles in the transcriptional regulation in plants (Wang et al. 2020). They found the LTR and MBS cis-elements in many PtrAHL promoters, indicating their role under low temperature and drought stress. These findings suggest that AHLs may play an important role in response to abiotic stress and phytohormone treatment in *P. trichocarpa* (Wang et al. 2021a, b).

Expression profiling of OsAHLs

The AHL genes are engaged in plant growth, floral transition, and abiotic and biotic stress responses (Zhao et al. 2014, 2013; Jin et al. 2011). AHL2, AHL4, AHL7, AHL8, and AHL11 were all upregulated in the embryo, but only at extremely low levels in the endosperm, leaf, anther, and pistil. AHL1, AHL7, AHL9, and AHL10 were shown to have widespread expression activity in several organs, with the pistil being the most strongly expressed, implying that these genes play a unique role in flower development. AHLs in each clade had similar expression patterns, although the expression patterns of AHLs in one monophyletic group differed from those of AHLs in the other.

In plants, water is essential for normal physiological operations like proliferation, development, and breeding (Fang and Xiong 2015). Woody plants devised a number of several techniques to overcome and endure drought, such as reprogramming expression levels (Debnath et al. 2011). In addition, cis-elements, particularly the MBS cis-element, have a critical function under drought stress. The MBS cis-element was used to identify 14 OsAHL genes in this investigation. Plant roots are preliminary to recognizing drought stress (Lynch 1995). Drought impacts a variety of processes in the leaf, comprising the closing of stomata to reduce the loss of water (Qi et al. 2020). As a result, during drought stress, we looked at the expression analysis of 26 OsAHL genes in *O. sativa* roots and leaves. Drought stress enhanced the expression of OsAHL4, 5, 23, and 24 genes in roots, showing that they play a role in drought response.

Comparing these findings with previous knowledge, the expression patterns of AHL genes in O. sativa align with their roles in plant growth, floral transition, and stress responses observed in other plant species. The specific expression patterns observed in different organs provide insights into the potential functions of AHL genes in organ development and stress-related processes. To further investigate the regulatory mechanisms underlying the expression patterns of AHL genes, subsequent studies could focus on identifying the transcription factors and signaling pathways involved in their regulation. Additionally, functional characterization of the identified cis-elements, such as the MBS cis-element, can shed light on their precise roles in drought stress responses. Furthermore, investigating the interaction networks of AHL proteins in response to drought stress and other environmental cues would be valuable. Protein–protein interaction studies, can elucidate the molecular mechanisms underlying the functions of AHL proteins and their interactions with other regulatory components.

Overall, by further exploring the expression profiles of AHL genes and investigating their regulatory mechanisms, we can deepen our understanding of their roles in plant development, stress responses, and potentially uncover novel targets for improving crop tolerance to drought and other environmental stresses.

Conclusion

Previous studies have revealed that AHL genes are a significant role in plant development and stress responses (both biotic and abiotic). This research looked at the AHL gene family in the genome of *O. sativa*. In conclusion; our research identifies 26 AHLs in *O. sativa* for the first time. The phylogeny, conserved features, and protein interaction network are all examined in detail, providing useful information regarding AHLs. Furthermore, based on promoter cis-element analysis, a systematic method was employed to explore OsAHL genes' involvement in response to stress and phytohormones. Our findings will aid in the identification of possible AHL genes and give a conceptual platform for future study into AHL gene activity in *O.sativa* under ABA treatment and drought stress. Based on phylogenetic and synteny investigations, AHLs in rice were determined to be identical to those in *A. thaliana*. AHL genes are found in abundance in both rice and Arabidopsis. According to sequencing analysis, segmental duplications were the main drivers of AHL family formation, showing that AHLs proliferated with distinct WGD in rice. It supports research that found and analyzed the whole AHL gene family in rice. The ratios of non-synonymous (Ka) to synonymous (Ks) substitution rates between paralogous gene pairs demonstrated that the *O. sativa* AHL genes had undergone different selections over time. Further, RNA-seq transcriptome analysis revealed that the majority of Clade-A AHLs expressed predominantly in the embryo, whereas Clade-B AHLs expressed predominantly in the pistil, implying that functional divergence may have happened because AHLs within each clade had comparable expression patterns. Our findings served as a starting point for additional functional research into these putative AHL proteins.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s13205-023-03666-0.

Acknowledgements The authors acknowledge the Sardar Vallabhbhai Patel University of Agriculture and Technology (Government of Uttar Pradesh), Meerut, India, as well as Noida International University, Gautam Budh Nagar, India for the required facilities.

Author contributions AK and SS conception or design of the work and wrote the main manuscript text. SS and AM data analysis, interpretation, and prepared figures. All authors reviewed the manuscript.

Funding The author (s) received no financial support for the research and publication of this research article.

Data availability Not Applicable.

Declarations

Conflict of interest The authors declare no conflict of interest.

Compliance with ethical standard This article does not contain any studies involving human participants and/or animals performed by any of the authors.

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